Reverse Genetics of AN8455 in *Aspergillus nidulans* by Af-pyrG+ Knockout-Cassette Replacement

Author: Yuewei.Xu Student ID:
Department of Biosciences and Bioinformatics, Xi'an Jiaotong-Liverpool University, China

Abstract

- Applied an **Af-pyrG+ knockout-cassette mediated reverse-genetics workflow** to analyze the *A. nidulans* gene **AN8455**.
- **Diagnostic PCR** supported the successful replacement of the endogenous AN8455 ORF by the Af-pyrG+ cassette in transformants.
- Observed **off-target bands** were interpreted as **non-specific amplification** under improper PCR conditions.

Introduction

- **Reverse genetics:** characterize genes through phenotypic analysis after targeted gene disruption.
- **Knockout cassette:** strategy for perturbation of target gene through homologous recombination.
- **Af-pyrG+:** encode terminal enzyme in uridine 5′-phosphate biosynthesis → only successful transformants can survive without exogenous uridine and uracil.
- **nkuA:** facilitate homologous recombination.

Methods & Workflow

Bioinformatics Analysis → Knockout Cassette Design → Transformation & Purification → Diagnostic PCR → Phenotype Analysis (Future)

Results & Analysis

1. Bioinformatics Analysis

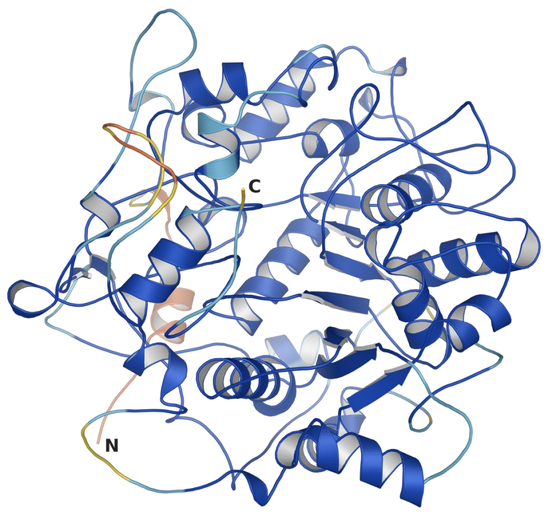


Fig. 1 Predicted AN8455 encoding protein structure – predicted to be **intracellular, non-secreted, non-transmembrane**, and containing a **DUF3405 domain**.

2. Genomic Annotation & Cassette Design

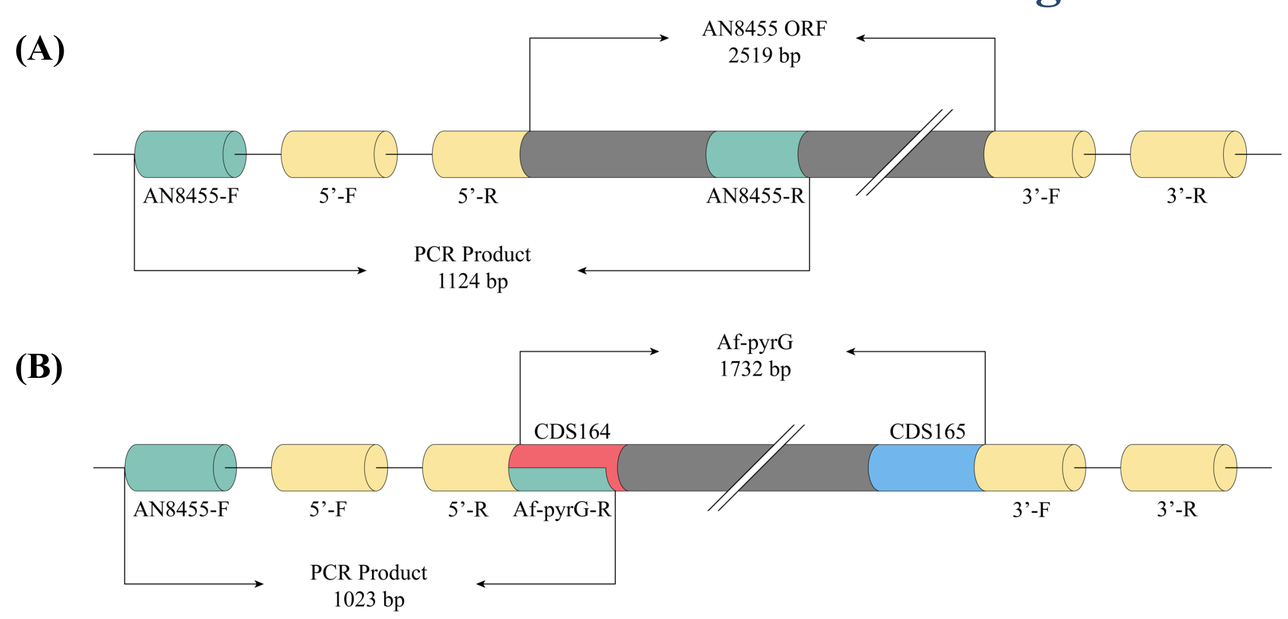


Fig2. schematic diagram for two structures.
(A) Genomic organization of the **AN8455 locus**.
(B) Predicted **Af-pyrG replacement structure**.

3. Transformation & Purification

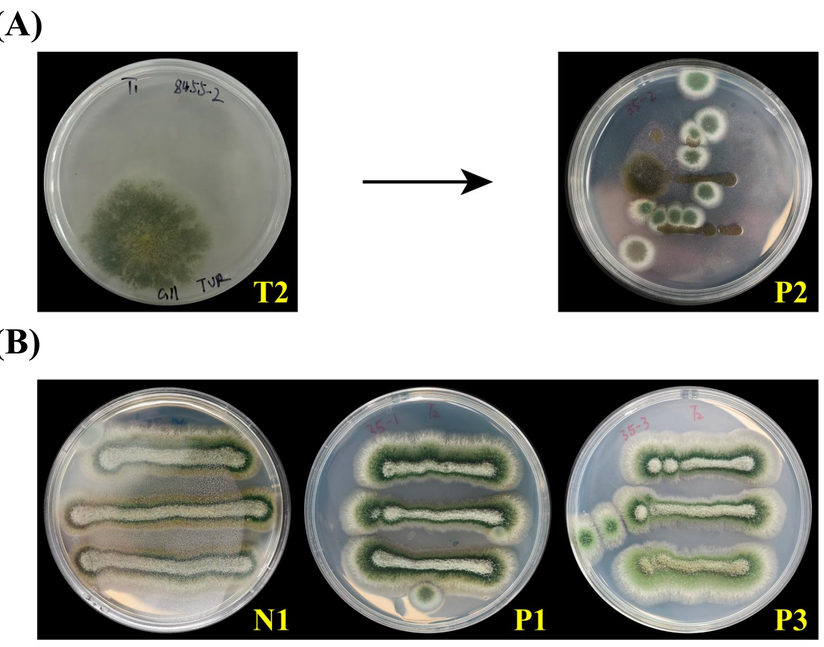
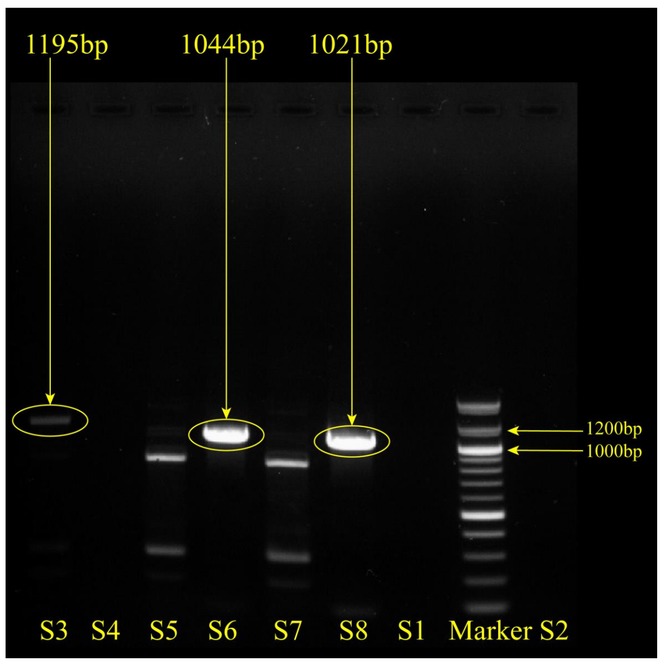


Fig 3. (A) Putative transformant **T2 from Group11**, which was purified into **P2**.
(B) Three strains used for diagnostic PCR. **N1:** negative group contained original strain. **P1 & P3:** positive groups contained putative transformants.

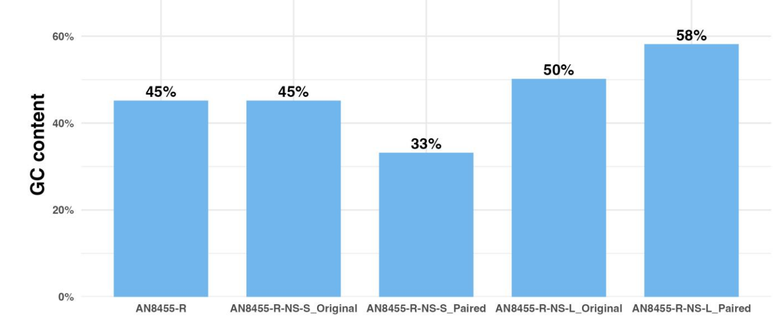
4. Diagnostic PCR

Fig 4. Gel photos, showing that the original strain control yielded a **1195** bp band, close to the predicted 1124 bp product (**+6.93%**), whereas the two putative transformants yielded **1044 bp** and **1021 bp** bands, both consistent with the expected 1023 bp transformant product (**+2.05% and −0.20%**).



✓**Transformation succeeded**

5. Off-target Bands Analysis



✓**Inference: two non-specific annealing sites**

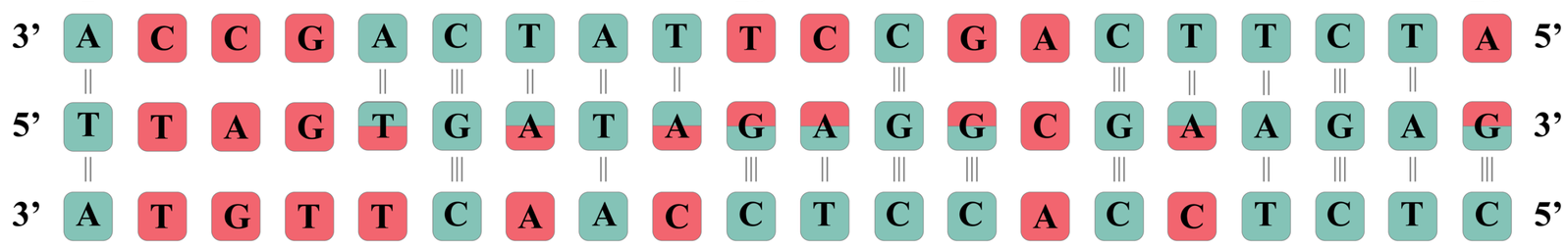


Fig 5. Comparison of Two putative non-specific annealing sites (AN8455-R-NS-S/L) to AN8455-R. (A). **GC contents** comparison. (B) **Specific alignment patterns** between AN8455-R (middle) and two putative non-specific annealing site (up & down). **The alignment was especially good in 3' of AN8455-R**.

Conclusion

- ✓ **Successfully constructed knockout cassette design.**
- ✓ **Successful transformation: validated by diagnostic PCR.**
  - ✓ **original strain: 1195 bp (predicted 1124 bp).**
  - ✓ **two putative transformants: 1044 & 1021 bp (predicted 1023 bp).**
  - ✓ **off-target bands: two non-specific amplification products.**
- ✓ **AN8455 encodes a conserved intracellular protein with DUF3405.**

Future Works

- **Phenotype-level functional analysis:**
  - Phenotypic comparison: hyphal morphology, growth kinetics, etc.
  - Complementary strategy: forward genetics + reverse genetics.
  - Computational & simulation based approaches.
- **Improve transformation efficiency:** CRISPR/Cas9-based gene editing.
- **Off-target bands validation:** bands sequencing / temperature gradient-PCR.